A

SEMINAR REPORT

seminar Report on

E-Waste Management & Recycling Portal

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THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR: 2026 – 2027 DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

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ABSTRACT

Electronic waste, commonly called e-waste, includes discarded electrical and electronic products such as mobile phones, computers, laptops, printers, televisions, cables, chargers and household appliances. Rapid technological change and frequent replacement of devices have increased the amount of electronic equipment that becomes unwanted. If such products are mixed with ordinary waste, dismantled carelessly or sent to unsuitable facilities, hazardous substances may affect human health and the environment. At the same time, e-waste contains reusable components and valuable materials that can be recovered through appropriate processes.

The proposed E-Waste Management & Recycling Portal is a web-based application intended to make responsible disposal information easier to access. It provides educational information about e-waste, explains safe handling practices, displays collection-centre or recycler details, and allows a user to submit information about an item that needs disposal. The request can be stored in a database and reviewed by an administrator, who can update its status. The portal therefore provides a structured point of contact between users and available recycling services.

The system can be developed using HTML and CSS for the user interface, PHP for server-side processing and MySQL for data storage. XAMPP can provide a local Apache and MySQL environment during development and testing. Main functions include viewing awareness content, browsing service information, submitting a request, checking its progress and maintaining records through an administrator section. Input validation and access control are important to protect the information entered by users.

The expected outcome is an academic prototype that demonstrates how a web portal can support awareness, record keeping and coordination for electronic waste management. It does not itself perform physical collection or recycling, and the usefulness of the portal depends on the accuracy of listings and participation of real service providers. Proper environmental results require suitable collection channels and responsible treatment of collected devices.

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CHAPTER 1

INTRODUCTION

Electronic products have become an essential part of everyday life. Mobile phones, computers, televisions, printers, routers and other electronic devices are used in homes, educational institutes, offices, hospitals and industries. These products help people communicate, learn, work and access information quickly. However, technology develops rapidly, and devices are frequently replaced because of damage, changing needs or the availability of newer models. The discarded products are commonly known as electronic waste or e-waste.

E-waste is different from ordinary household waste because it may contain a mixture of plastics, glass, metals, batteries and electronic components. Some materials can be recovered and reused, while certain components need careful handling. If electronic products are broken apart or disposed of in an unsuitable manner, harmful substances may be released into the environment. Informal handling can also expose workers to unsafe conditions. Therefore, electronic waste should be collected, sorted and sent to appropriate facilities.

The E-Waste Management & Recycling Portal is proposed as an online information and request-management system. The portal is intended to help users understand the importance of responsible disposal and find available collection options. Instead of searching across many sources, a visitor can view instructions, browse listed collection centres and submit a request with basic details about the item. An administrator can review the submitted details and maintain the service information.

1.1 Need for the Project

Many people keep old devices at home because they do not know where they should be taken or whether a collection centre accepts a particular item. Information about recycling services may be spread across websites, notices or local contacts. A simple portal can make this information easier to locate and can provide a clear process for registering an item. It can also support educational institutes and offices that want to encourage systematic disposal of unused electronic equipment.

1.2 Aim and Objectives

The aim is to design a user-friendly web portal that supports awareness and organises information about e-waste collection and recycling. Objectives include explaining e-waste hazards, presenting disposal guidance, maintaining collection-centre details, accepting item requests, providing request-status information and allowing an administrator to manage records.

CHAPTER 2

LITERATURE REVIEW

A review of existing approaches helps identify the methods commonly used to collect electronic waste and communicate disposal instructions. E-waste programmes generally rely on authorised collection points, manufacturer take-back arrangements, local drives, recycling facilities and awareness campaigns. Online information systems can complement these approaches by helping people find service details and understand the steps required before handing over an item.

1. Collection-centre directories

Directories provide a list of locations where selected electronic products can be delivered. Useful details include address, contact information, operating hours, accepted categories and any preparation instructions. The value of a directory depends on the accuracy and regular updating of its records. If details are outdated, users may travel to a centre that no longer accepts the item.

2. Manufacturer take-back programmes

Some manufacturers and retailers provide take-back or exchange programmes for selected products. These programmes may be convenient because users can return old equipment through an established channel. However, programme availability and accepted product types vary. A user must check the applicable terms before relying on a particular service.

3. Community collection drives

Schools, colleges, offices and local organisations may arrange collection drives to gather unused devices. Such activities can increase awareness and collect multiple items at one time. Their limitations include the need for coordination, advance announcements, temporary collection arrangements and a suitable downstream recycling partner.

4. Awareness websites

Educational websites explain what e-waste is, why unsafe disposal is harmful and how equipment can be reused or recycled. A portal can combine awareness material with a practical request form so that information is linked to an action. The content should use clear language and avoid suggesting that every listed service is available in every locality.

LITERATURE REVIEW (CONTINUED)

5. Online service request systems

Request-based systems allow users to provide item details and contact preferences through a form. Records can be reviewed by staff, assigned a status and retained for follow-up. This approach makes the request process easier to monitor than informal messages alone. The system should explain that submitting a request does not guarantee collection until a service provider confirms the arrangement.

6. Database-driven web applications

A database-backed website separates page presentation from stored records. User requests, item categories and collection-centre listings can be stored in organised tables. This enables administrators to add, update or remove records without editing every page manually. Validation, suitable access permissions and backups are important parts of maintaining data quality.

7. Existing practice and identified gap

Many existing e-waste approaches address one part of the process, such as education, location information or collection. Users may still need to visit several sources to understand the correct process and contact an appropriate service. A small integrated portal can bring awareness, centre details and request submission into one interface. The proposed system is designed as an academic prototype and does not replace government guidance or verified recycling operations.

8. Summary of review

The review suggests that convenient information, trustworthy listings and a transparent request process are useful for improving user experience. The portal can contribute by providing one organised interface and a manageable record system. Its success depends on regular content review, cooperation with service providers and responsible handling of personal information. The system should also clearly separate sample demonstration data from verified service listings.

LITERATURE REVIEW (CONTINUED)

The approaches reviewed above can be compared on the basis of their main purpose, information required and practical limitations. No single online tool can ensure safe recycling by itself; digital information must connect users with suitable physical collection and processing arrangements.

Sr. No.	Approach	Main purpose	Limitations
1	Collection-centre directory	Shows locations and accepted items	Listings may become outdated
2	Take-back programme	Provides return route for selected products	Depends on provider rules and coverage
3	Community collection drive	Collects multiple items in a planned event	Needs organisation and a downstream partner
4	Awareness website	Explains hazards and safe disposal	May not offer a direct request workflow
5	Online request system	Records user details and tracks progress	Submission alone cannot guarantee pickup
6	Database-driven portal	Centralises records and administration	Needs security, backups and maintenance

Table 2.1: Literature Review

The comparison indicates that a portal combining awareness, service listings and request tracking can improve access to information. The project focuses on these features while recognising that actual collection and recycling must be carried out by suitable service providers.

Table No. 2.1: Literature Review (Continued)

Sr. No.	Approach type	Information / technology	Key objective	Limitation
1	Centre directory	Web pages; database records	Help users locate drop-off options	Needs frequent verification
2	Take-back scheme	Provider website and contact process	Return products through an established channel	Coverage differs by product and area
3	Collection campaign	Registration lists and event notices	Collect items from a group of users	Usually available only at planned times
4	Awareness material	Text, images and guidance pages	Explain environmental risks and safe actions	May not help users contact a service
5	Request portal	HTML forms, PHP and MySQL	Capture item details and provide status	Requires administrator follow-up
6	Service management	Admin dashboard and records	Maintain listings and review requests	Requires login security and backups

Interpretation

The reviewed methods provide different kinds of support. Directories help with discovery, take-back schemes and collection drives provide physical routes, and awareness material explains why responsible disposal matters. A web portal can connect these elements through organised information and a request record. Accuracy and accountability remain essential: listings should be checked, unconfirmed services should not be presented as guaranteed, and request statuses should reflect the latest available information.

LITERATURE REVIEW: COMPARATIVE NOTES

The following points describe common strengths and constraints that should be considered while designing an e-waste information portal.

Information quality

The portal should display clear service names, locations, accepted item categories and contact methods where known. Each listing should have a review date or maintenance process. When a service has not been verified, the portal should state that limitation instead of implying official approval.

Usability

A simple layout, readable headings and short instructions can help users of different experience levels. The form should request only the details needed to understand an item and respond to the user. Error messages should explain what needs correction instead of simply rejecting the submission.

Privacy and security

Contact details and request descriptions should be visible only to authorised administrators where appropriate. Server-side validation, parameterised database queries and protected administration pages reduce common risks. The system should avoid collecting unnecessary sensitive information and should not publish personal request data publicly.

Reliability and maintenance

A portal requires regular maintenance even if its first version is small. Database backups, correction of broken links, removal of outdated listings and testing after changes are important. If collection partners or external services are added later, their participation and data accuracy should be confirmed.

Conclusion of review

The project applies basic web development to a practical information problem. It is intentionally limited to information display and request management so that the prototype can be developed and demonstrated within an academic project schedule.

LITERATURE REVIEW: DESIGN IMPLICATIONS

Based on the review, the proposed portal should be organised around the user's journey: understand the issue, find a suitable option, submit item information and check the recorded status. The administrator's journey should include reviewing requests, updating collection-centre details and maintaining accurate content.

Requirement	Design implication
Easy discovery	Use clear navigation and a dedicated collection-centre section
Clear disposal guidance	Provide concise instructions and safety notes
Structured requests	Use a form with required fields and validation
Status visibility	Assign each request a unique reference and current status
Manageable records	Use relational tables and an administrator interface
Trustworthy information	Review service listings and label unverified data
Responsible data use	Limit collection of personal details and restrict access

Table 2.1: Summary of requirements derived from review

These design implications form the basis for the system description in the next chapter. They also establish a realistic boundary for the academic prototype: the application can organise information and requests, while the actual collection and treatment process remains the responsibility of the relevant service provider.

CHAPTER 3

INFORMATION/CHAPTERS

1. Overview of E-Waste Management & Recycling Portal

The E-Waste Management & Recycling Portal is a website concept that helps users access electronic waste information and organise disposal requests. The portal is placed between the person seeking a disposal option and the information maintained by the project administrator. It does not physically collect or recycle equipment. Instead, it provides a structured digital interface through which users can learn about e-waste, find listed options and record a request.

2. Main objectives of the system

The system aims to make disposal guidance easier to find, keep service information organised, capture item requests in a database and provide a simple way to review request progress. It should be understandable for users who have limited technical knowledge. The administrator should be able to manage the records without directly changing the website source code for every update.

3. Types of electronic waste

Examples include mobile phones, tablets, laptops, desktop computers, monitors, keyboards, printers, cables, chargers, routers, televisions and selected household appliances. Batteries and damaged equipment may require specific handling instructions. The portal should advise users to follow local rules and the instructions of the receiving facility rather than dismantling devices at home.

4. General e-waste handling process

A responsible process begins when an unwanted device is identified. If the device can be safely repaired or reused, that option may be considered first. Otherwise, the user checks accepted-item information, prepares the item according to the receiving facility’s instructions and uses an appropriate collection or drop-off channel. The receiving service is responsible for sorting, reuse, dismantling or recycling under its applicable procedures.

4. WORKING PRINCIPLE

The portal workflow consists of a series of simple steps that connect the user interface with the database and administrator functions.

Step 1: Open the website. The user visits the home page and reads the introductory information about e-waste management.

Step 2: Review guidance. The user checks which kinds of electronic items are covered and reads the safe-handling instructions.

Step 3: Find a service option. The user opens the collection-centre or recycler information page and checks available details.

Step 4: Enter item information. If the user wants to register an item, a form is completed with the item category, short description, quantity if required and suitable contact or preference fields.

Step 5: Validate the form. The website checks required fields and the server validates submitted values before processing them.

Step 6: Save the request. The application stores the valid request in the database and provides a reference or confirmation message.

Step 7: Administrator review. An authorised administrator reviews the new request and updates its status when more information becomes available.

Step 8: Check status. The user can refer to the request number or status page to view the latest recorded progress. A status should not be interpreted as proof of completed recycling unless that information is confirmed by the service provider.

Fig. 3.1: User Request Workflow

User	Portal	Administrator
Reads guidance	Displays pages and validates input	Maintains listings
Submits item request	Stores request in database	Reviews pending request
Checks status	Shows latest recorded status	Updates request status

WORKING PRINCIPLE (CONTINUED)

Step 9: Follow safe instructions. The user follows the collection facility’s instructions for packaging, batteries, damaged devices and delivery. The portal should not encourage unsafe dismantling or disposal.

Step 10: Maintain records. The administrator corrects outdated information, closes requests when appropriate and ensures that only authorised staff can access administrative functions.

5. Data flow

The user interacts with the presentation layer through a browser. Form data is sent to PHP scripts, which perform validation and use database queries to store or retrieve records from MySQL. The result is returned to the browser as a confirmation, listing or status message. The administrator section uses the same application and database but must require authentication and permission checks.

6. Input and output

Inputs may include the item category, brief item description, quantity, preferred option and necessary contact details. Outputs may include awareness information, collection-centre listings, validation messages, a request reference and a status update. The system should avoid displaying private contact information on public pages.

7. Error handling

If required fields are missing, the form should identify them. If a database operation fails, the user should receive a clear message that the request could not be completed, without being shown database credentials or internal error traces. The application should prevent duplicate or malformed submissions where practical and should record only valid data.

8. User roles

A visitor can read public information and browse service listings. A user submitting a request may provide the information required for follow-up. An administrator can add or edit service details and review requests. Role separation prevents an ordinary visitor from accessing management functions.

5. SYSTEM ARCHITECTURE AND MAIN COMPONENTS

The proposed architecture contains a browser-based user interface, server-side PHP logic and a MySQL database. These components work together to present information, process requests and maintain records.

Component	Responsibility
User interface	Displays home, awareness, service, request and status pages
Request handler	Validates form data and processes user submissions
Database layer	Stores users where needed, item requests, categories and service listings
Administrator module	Authenticates authorised staff and manages records
Status module	Retrieves and displays the latest recorded request state
Security controls	Validate inputs, restrict access and protect database operations

Fig. 3.2: System Architecture of E-Waste Portal

The architecture is suitable for a prototype because it uses common web technologies and can be run locally. A production deployment would require stronger operational controls, secure hosting, backup arrangements and confirmed service-provider processes.

6. MODULES AND EXAMPLE OF PORTAL USE

6.1 Home and awareness module

The home page introduces the purpose of the portal and provides navigation to the main sections. The awareness module explains what electronic waste includes, why it should be handled responsibly and what users can do before disposal. Text should be concise and practical.

6.2 Collection-centre module

This module displays the details available for collection points or recycling services. A listing may include name, location, accepted item categories and contact details when verified. The page should be updated when service information changes and should distinguish sample records from confirmed listings.

6.3 Request module

The request form captures the information needed to understand the item and arrange follow-up. After validation, the application saves a record and provides a confirmation. The system should explain that the request is pending until an appropriate service confirms the next step.

6.4 Administrator module

The administrator can review submitted requests, update their status, add or correct collection-centre records and remove obsolete information. The administrator area should be protected by authentication, and database operations should use parameterised queries.

6.5 Example

A household has an unused laptop and charger. The user visits the portal, reads the instructions and checks the listed service options. The user enters the item details in the request form and submits it.

The application validates the form, stores the record and shows a reference number. The administrator reviews the request and updates the status after checking the available service. The portal communicates the recorded status but does not claim that collection has occurred until it is confirmed.

6.6 Technologies used

HTML defines page structure, CSS controls presentation, PHP processes requests and connects to MySQL, and XAMPP provides a local development environment. A browser is used for testing the interface. These tools are appropriate for demonstrating basic dynamic website and database concepts.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages

1. Awareness: The portal explains the meaning of e-waste and encourages users to think about responsible disposal.
2. Centralised information: Users can access guidance and available collection details from one website instead ofsearching multiple sources.
3. Simple request submission: A form provides a structured method for recording item information and requestingfollow-up.
4. Better record management: Requests and service listings can be stored in a database and reviewed by anadministrator.
5. Status visibility: A request reference and current status can help users understand whether a request is pending or hasbeen updated.
6. Expandability: Additional languages, map views, notifications or reporting features may be added later if the projectrequirements and resources permit.

4.2 Disadvantages

1. Dependence on current information: Incorrect or outdated collection-centre details can mislead users.
2. Internet access: Users need a device and internet connection to access the online portal.
3. No physical collection by the software: A website cannot itself pick up, transport or recycle electronic products.
4. Maintenance needs: Administrators must review listings, manage database records and correct errors.
5. Security risks: Poor validation or weak administrator authentication could expose submitted information or allowunauthorised changes.

4.3 Risk reduction

The portal should maintain accurate service information, use clear wording about availability, restrict administrative access, validate inputs and protect user data. It should not guarantee collection or recycling unless a service provider has confirmed it. Regular testing and backups can improve reliability.

CHAPTER 5

CONCLUSION

Electronic waste is an important environmental concern because electronic products are replaced frequently and can contain materials that require suitable handling. Responsible management depends on awareness, convenient access to collection options and the use of appropriate treatment channels. Users may find it difficult to identify the correct place for an unwanted device when service information is scattered or unclear.

The proposed E-Waste Management & Recycling Portal offers a central online interface for awareness content, collection-centre information and request management. It demonstrates how HTML, CSS, PHP and MySQL can be combined to build a dynamic website that stores item requests and allows an administrator to maintain records. The project also highlights the importance of input validation, access control and regular information updates.

The portal is an academic prototype and does not independently perform collection or recycling. Its effectiveness depends on whether the information is accurate and whether suitable collection services are available. A request submission is only a record of user interest; actual pickup and processing must be confirmed by the responsible service provider.

5.1 Future scope

Future improvements may include a map-based collection-centre search, multiple-language support, email or SMS notifications, appointment requests, a verified recycler profile process, an administrator report dashboard and improved mobile layouts. Integration with external organisations should be introduced only after service details and operational responsibilities are confirmed.

In conclusion, the project provides a useful demonstration of how a web portal can support public awareness and organise information related to e-waste. With accurate data, responsible administration and cooperation from appropriate service providers, such a system may help users take more informed steps toward responsible disposal.

CHAPTER 6

PROJECT INITIATION

6.1 Description of Problem Statement

The increasing use of electronic devices results in a continuing stream of unwanted equipment. Users may store old phones, computers, chargers and other products because they are uncertain about safe disposal or do not know which collection options are available. When devices are discarded with ordinary rubbish or handled without appropriate precautions, environmental and safety risks may arise. Valuable components may also be lost instead of being directed to reuse or recovery.

Information about collection facilities can be distributed across different websites, notices and local contacts. Users may not know whether a facility accepts a particular item, what preparation is required or whom to contact. Informal requests may be difficult to track, and administrators may lack one organised place to maintain requests and service listings.

The proposed project addresses this information and coordination problem through a web-based portal. It provides educational content, a directory of available options and a structured request form. Requests can be saved in a database and reviewed by an administrator. The project aims to make the process clearer, while recognising that the website itself cannot guarantee physical collection or recycling.

Problem statement: To design and develop a simple web portal that provides e-waste awareness, organises available collection information and supports registration and tracking of disposal requests through a database-driven interface.

6.2 Scope and Objectives

The scope includes a home page, e-waste awareness pages, collection-centre information, a request submission form, a request-status view and an administrator module for managing records. The prototype can be developed and tested in a local XAMPP environment. Actual pickup scheduling, payment processing, logistics and certification of recycling outcomes are outside the basic project scope unless separately implemented and verified.

Objectives of the project

1. To provide simple information about electronic waste and responsible disposal.
2. To organise collection-centre or recycler information in a single interface.
3. To allow users to submit basic details about an unwanted electronic item.
4. To store valid requests in a MySQL database.
5. To allow an administrator to review requests and update recorded status.
6. To apply basic validation and access-control practices.
7. To demonstrate the use of HTML, CSS, PHP, MySQL and XAMPP in a practical web application.

6.3 Stakeholders

Users and households need clear instructions and convenient access to service information. Students, schools and offices may use the portal to organise awareness activities or item requests. Collection centres and recycling service providers may be represented in the directory where details have been checked. The project administrator maintains content and processes requests. Teachers and evaluators assess the design, implementation and demonstration of the academic prototype.

Stakeholder needs

Users need a simple interface and understandable status messages. Administrators need organised records and manageable update functions. Service providers need accurate descriptions and should not be represented as participating without confirmation. Evaluators need a clear explanation of the problem, scope, technologies, limitations and expected outcome.

6.4 Platform / Equipment / Resources

1. Platform

The portal can be developed on a Windows or Linux computer using a standard web browser and a local web server.

XAMPP provides Apache and MySQL services for local testing. The source code can be written in a code editor such as Visual Studio Code or another suitable editor. The prototype does not require specialised hardware.

2. Hardware requirements

A computer or laptop with sufficient memory and storage for a browser, editor and local server is adequate for student development. A stable power supply and backup of source files and database exports are recommended. Internet access is useful for research and for checking official information, but local testing can be performed on the same computer. **3. Software requirements**

Resource	Purpose
Operating system	Runs the development tools
XAMPP / Apache	Provides local web server environment
PHP	Processes forms and server-side logic
MySQL / MariaDB	Stores requests and service records
HTML and CSS	Builds and styles web pages
Code editor	Writes and maintains project files
Web browser	Displays and tests the portal

Table 6.1: Resources 4. Basic system requirements

The functional requirements are to view awareness pages, browse service information, submit a request, store records, review requests and update status. Non-functional requirements include readable design, reasonable response time, input validation, protected administrator access, maintainable code and regular backups. These requirements provide a basis for testing the prototype.

CHAPTER 7

SEMINAR PRESENTATION

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Abstract

Electronic waste (e-waste) includes discarded phones, computers, laptops, printers, televisions, chargers and household appliances. Rapid technological change has increased the amount of unwanted equipment. If it is mixed with ordinary waste or handled carelessly, hazardous substances can harm health and the environment, although e-waste also contains reusable components and valuable materials.

The proposed E-Waste Management & Recycling Portal is a web-based application that makes responsible disposal information easier to access. It offers awareness content and safe-handling guidance, displays collection-centre details and lets users submit a disposal request that an administrator can review and update. It uses HTML and CSS, PHP and MySQL, tested locally on XAMPP.

The expected outcome is an academic prototype that supports awareness, record keeping and coordination. The portal does not perform physical collection or recycling, and its usefulness depends on accurate listings and real service providers.

1. Introduction

- E-waste means discarded devices such as phones, laptops, printers and chargers.
- Frequent replacement of devices keeps increasing the amount of e-waste.
- Careless disposal can release harmful substances and waste reusable materials.
- Users often do not know where an old device can be handed over.
- Aim: a user-friendly portal that gives guidance, lists collection centres and accepts disposal requests.



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2. Literature Review

Sr. No.	Approach Type	Information / Technology	Key Objective	Limitation
1	Centre directory	Web pages; database records	Help users locate drop-off options	Needs frequent verification
2	Take-back scheme	Provider website and contact process	Return products through an established channel	Coverage differs by product and area
3	Collection campaign	Registration lists and event notices	Collect items from a group of users	Usually available only at planned times

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Sr. No.	Approach Type	Information / Technology	Key Objective	Limitation
4	Awareness material	Text, images and guidance pages	Explain environmental risks and safe actions	May not help users contact a service
5	Request portal	HTML forms, PHP and MySQL	Capture item details and provide status	Requires administrator follow-up
6	Service management	Admin dashboard and records	Maintain listings and review requests	Requires login security and backups

6

Literature Review: Design Implications

Requirement	Design implication
Easy discovery	Clear navigation and a dedicated collection-centre section
Clear disposal guidance	Concise instructions and safety notes
Structured requests	A form with required fields and validation
Status visibility	A unique reference and current status for each request
Manageable records	Relational tables and an administrator interface
Trustworthy information	Review service listings and label unverified data
Responsible data use	Limit collection of personal details and restrict access

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Literature Review: Key Observations

- **Information quality:** Show clear service names, locations, accepted items and contacts; review listings and label unverified services.
- **Usability:** Simple layout and short instructions; the form asks only for needed details and gives clear error messages.
- **Privacy and security:** Contact details visible only to authorised administrators; server-side validation, parameterised queries and protected admin pages.
- **Reliability and maintenance:** Regular backups, fixing broken links, removing outdated listings and testing after changes.
- **Conclusion of review:** The project applies basic web development to a practical information problem and stays limited to information display and request management, so it can be demonstrated within an academic schedule.

3. Information about the Seminar Topic

- **Overview:** A website concept that helps users access e-waste information and organise disposal requests. It does not collect or recycle equipment itself.
- **Technology:** HTML, CSS, PHP and MySQL, with XAMPP (Apache and MySQL) for local testing.
- **Main objectives:** Make disposal guidance easy to find, keep service information organised, capture item requests and let the administrator manage records without editing code.
- **Types of e-waste:** Mobiles, tablets, laptops, desktops, monitors, printers, cables, chargers, routers, televisions and some household appliances.
- **Handling process:** Identify the unwanted device, consider repair or reuse, check accepted items, prepare it as the facility instructs and use an appropriate drop-off channel. The receiving service sorts, reuses, dismantles or recycles.

3.1 Working Principle

1. **Open the website:** user reads the home page and introduction.
2. **Review guidance:** check covered items and safe-handling instructions.
3. **Find a service option:** open the collection-centre page and check details.
4. **Enter item information:** category, description, quantity and contact preference.
5. **Validate the form:** required fields and server-side checks on all values.
6. **Save the request:** stored in the database with a reference message.
7. **Administrator review:** an authorised admin reviews it and updates the status.
8. **Check status:** user views the latest recorded progress using the request number.
9. **Follow safe instructions:** follow the facility rules for packaging, batteries and delivery.
10. **Maintain records:** admin corrects outdated information and closes requests.

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3.2 User Request Workflow

User	Portal	Administrator
Reads guidance	Displays pages and validates input	Maintains listings
Submits item request	Stores request in database	Reviews pending request
Checks status	Shows latest recorded status	Updates request status

Example

A household with an unused laptop and charger reads the guidance, submits the item details and receives a reference number. The administrator reviews the request and updates its status after checking the available service.

A status is not proof of completed recycling until the service provider confirms it.

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3.3 System Architecture and Components

Component	Responsibility
User interface	Displays home, awareness, service, request and status pages
Request handler	Validates form data and processes user submissions
Database layer	Stores users, item requests, categories and service listings
Administrator module	Authenticates authorised staff and manages records
Status module	Retrieves and displays the latest recorded request state
Security controls	Validate inputs, restrict access and protect database operations

Browser-based interface → PHP logic → MySQL database
Suits a prototype because it uses common web technologies and runs locally.

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3.4 Modules of the Portal

- **Home and awareness:** introduces the portal, explains what e-waste is and provides navigation.
- **Collection-centre module:** shows name, location, accepted categories and verified contacts; sample records are kept separate from confirmed listings.
- **Request module:** the form captures item details; after validation the request is saved and a reference number is shown. It stays pending until a service confirms.
- **Administrator module:** reviews requests, updates status and edits centre records; protected by login and parameterised queries.
- **Technologies:** HTML (structure), CSS (design), PHP (processing and MySQL connection), MySQL (storage) and XAMPP (local server).

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3.5 Data Flow, Input/Output and Roles

- **Data flow:** browser → PHP scripts (validation) → MySQL; the result returns as a confirmation, listing or status message. The admin area needs authentication.
- **Inputs:** item category, short description, quantity, preferred option and necessary contact details.
- **Outputs:** awareness information, centre listings, validation messages, request reference and status updates.
- **Error handling:** missing fields are identified; database failures show a clear message without credentials or internal traces; duplicate or malformed submissions are prevented.
- **User roles:** a visitor reads information and browses listings; a user submits a request; an administrator adds or edits service details and reviews requests.

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4. Advantages and Disadvantages

Advantages

- Awareness: explains e-waste and encourages responsible disposal.
- Centralised information: guidance and centre details on one website.
- Simple request submission through a structured form.
- Better record management: requests and listings stored in a database.
- Status visibility: users see whether a request is pending or updated.
- Expandability: languages, maps and notifications can be added later.

Disadvantages

- Depends on current information; outdated centre details can mislead users.
- Users need a device and an internet connection.
- Software cannot itself pick up, transport or recycle products.
- Administrators must maintain listings and database records.
- Weak validation or admin authentication creates security risks.

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5. Conclusion

E-waste is an important environmental concern because electronic products are replaced frequently and contain materials that need suitable handling. The E-Waste Management & Recycling Portal offers one online interface for awareness content, collection-centre information and request management.

It shows how HTML, CSS, PHP and MySQL can build a dynamic website that stores item requests and lets an administrator maintain records. It is an academic prototype: collection and recycling are carried out by confirmed service providers, not by the software.

Future Scope

Map-based centre search	**Multiple-language support**	**Email / SMS notifications**
Appointment requests	**Verified recycler profiles**	**Admin report dashboard**

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6. Project Initiation

The increasing use of electronic devices creates a continuing stream of unwanted equipment. Users may keep old phones, computers and chargers because they are unsure about safe disposal or do not know which collection options are available.

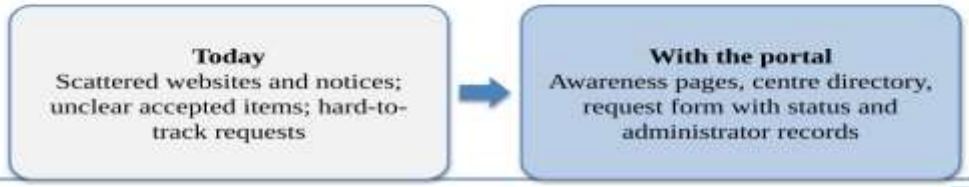
When devices are discarded with ordinary rubbish or handled without precautions, environmental and safety risks arise and valuable components are lost instead of being reused or recovered.

This project addresses the information and coordination problem through a web-based portal with educational content, a directory of options and a structured request form, while recognising that the website itself cannot guarantee physical collection or recycling.

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6.1 Problem Statement

- Unwanted devices pile up because users do not know where and how to dispose of them.
- Collection information is scattered across websites, notices and local contacts; informal requests are hard to track.
- **Problem statement:** to design and develop a simple web portal that provides e-waste awareness, organises collection information and supports registration and tracking of disposal requests through a database-driven interface.



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6.2 Scope and Objectives

Objectives

- Provide simple information about e-waste and responsible disposal.
- Organise collection-centre information in a single interface.
- Allow users to submit basic details about an unwanted item.
- Store valid requests in a MySQL database.
- Let an administrator review requests and update recorded status.
- Apply basic validation and access-control practices.
- Demonstrate HTML, CSS, PHP, MySQL and XAMPP in a practical web application.

Scope

- Home page, awareness pages, centre information, request form, status view and administrator module.
- Developed and tested in a local XAMPP environment.
- Out of scope: pickup scheduling, payments, logistics and certification of recycling outcomes.

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6.3 Stakeholders

- **Users and Households:** need clear instructions, a simple interface and understandable status messages.
- **Students, Schools and Offices:** use the portal to organise awareness activities or item requests.
- **Collection Centres and Recyclers:** listed once details are checked; need accurate descriptions and must not be shown as participating without confirmation.
- **Project Administrator:** maintains content and processes requests; needs organised records and manageable updates.
- **Teachers and Evaluators:** assess design, implementation and demonstration; need a clear explanation of problem, scope, technologies, limitations and expected outcome.

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6.4 Platform, Equipment and Resources

Platform:

- Windows or Linux computer with a standard web browser.
- XAMPP (Apache and MySQL) as the local web server.
- Code editor such as Visual Studio Code.

Hardware Requirements:

- Computer or laptop with enough memory and storage; no specialised hardware.
- Stable power supply; backup of source files and database exports.

Software Requirements:

- Operating system; XAMPP / Apache; PHP; MySQL / MariaDB.
- HTML and CSS; code editor; web browser.

System Requirements:

- Functional: view awareness pages, browse services, submit request, store records, review and update status.
- Non-functional: readable design, reasonable response time, input validation, protected admin access, backups.

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Bibliography

Books

- S. K. Garg — Environmental Engineering and study material on solid waste management.
- E. Balagurusamy — Programming with PHP and MySQL (introductory learning material).
- Learning resources on web development, database management systems and software engineering.

Online and Technical Resources

- Central Pollution Control Board (CPCB), Govt. of India — e-waste management guidance.
- MoEF&CC, Govt. of India — environmental rules and notifications.
- PHP Manual and MySQL Documentation — server-side development and databases.
- Apache Friends — XAMPP documentation and local setup.
- HTML and CSS learning documentation — page structure, styling and responsive layouts.

Project and Learning Material

- Course material for Seminar Presentation and Project Initiation (Course Code: 315003).
- Lab practice notes on dynamic website development using PHP and MySQL.

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1. Central Pollution Control Board (CPCB), Government of India. e-waste management information. https://cpcb.nic.in/
2. Ministry of Environment, Forest and Climate Change, Government of India. https://moef.gov.in/
3. Government of India. E-Waste (Management) Rules, 2022, and subsequent amendments (consult the official publication).
4. PHP Documentation Group. PHP Manual. https://www.php.net/manual/en/
5. Oracle. MySQL Reference Manual. https://dev.mysql.com/doc/
6. Apache Friends. XAMPP documentation and downloads. https://www.apachefriends.org/
7. World Wide Web Consortium (W3C). Web standards for HTML and CSS. https://www.w3.org/
8. Mozilla Developer Network (MDN). HTML, CSS and web form documentation. https://developer.mozilla.org/

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9. General learning resources on database normalisation, input validation and secure web application development.
10. Course notes and practical exercises for PHP-based dynamic websites and database connectivity.
11. HTML Living Standard. WHATWG. https://html.spec.whatwg.org/
12. CSS specifications and learning material. W3C. https://www.w3.org/Style/CSS/
13. OWASP Foundation. Guidance on input validation, authentication and secure web development. https://owasp.org/
14. MySQL documentation on database design, data types, queries and user privileges. https://dev.mysql.com/doc/
15. PHP documentation on form handling, sessions and database access. https://www.php.net/manual/en/

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Thank you

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1. S. K. Garg, Environmental Engineering and related study material on solid waste management.
2. E. Balagurusamy, Programming with PHP and MySQL: introductory learning material.
3. Learning resources on web development, database management systems and software engineering.

Online and Technical Resources

1. Central Pollution Control Board (CPCB), Government of India — e-waste management information and guidance.
2. Ministry of Environment, Forest and Climate Change, Government of India — environmental rules and notifications.
3. PHP Manual — reference for PHP language features and server-side development.
4. MySQL Documentation — reference for database design, queries and administration.
5. Apache Friends — documentation for XAMPP and local development setup.
6. HTML and CSS learning documentation for page structure, styling and responsive layouts.

Project and Learning Material

1. Course material for Seminar Presentation and Project Initiation (Course Code: 315003).
2. Laboratory practice and notes on dynamic website development using PHP and MySQL.
3. General educational resources on electronic waste, reuse, recycling and safe handling.

The bibliography lists learning resources relevant to the topic and technologies. Before deployment beyond an academic demonstration, the project team should consult the latest official requirements and verify local service information.

REFERENCES

1. Central Pollution Control Board (CPCB), Government of India. Official website and information related to e-wastemanagement. https://cpcb.nic.in/
2. Ministry of Environment, Forest and Climate Change, Government of India. Official rules, notifications andenvironmental information. https://moef.gov.in/
3. Government of India. E-Waste (Management) Rules, 2022, and subsequent amendments or notifications asapplicable. Consult the official government publication for the current text.
4. PHP Documentation Group. PHP Manual. https://www.php.net/manual/en/
5. Oracle. MySQL Reference Manual. https://dev.mysql.com/doc/
6. Apache Friends. XAMPP documentation and download information. https://www.apachefriends.org/
7. World Wide Web Consortium (W3C). Web standards and guidance for HTML and CSS. https://www.w3.org/
8. Mozilla Developer Network (MDN). HTML, CSS and web form documentation. https://developer.mozilla.org/9. General learning resources on database normalisation, input validation and secure web application development.

10. Course notes and practical exercises for PHP-based dynamic websites and database connectivity.

The listed sources are provided as starting points for further study. Specific regulations, accepted-item rules and service availability should be verified from current official sources before the portal is used to guide real disposal decisions.

REFERENCES (CONTINUED)

11. HTML Living Standard. Web Hypertext Application Technology Working Group (WHATWG). https://html.spec.whatwg.org/
12. CSS specifications and learning material. World Wide Web Consortium (W3C). https://www.w3.org/Style/CSS/
13. OWASP Foundation. Guidance on input validation, authentication and secure web application development.https://owasp.org/
14. MySQL documentation on database design, data types, queries and user privileges. https://dev.mysql.com/doc/
15. PHP documentation on form handling, sessions and database access. https://www.php.net/manual/en/

Implementation Note

The portal should distinguish sample data used for demonstrations from verified collection services. It should not claim that a centre is authorised, that a pickup has been booked or that recycling has been completed unless this has been confirmed. Personal information should be limited to what is necessary for the request and protected through suitable access controls.

Future work may include checking local collection options, providing multilingual content, testing the site on mobile devices, improving error messages, reviewing security controls and preparing a clear process for maintaining service listings. Any real-world use should follow current official rules and the instructions of the receiving facility.

Expected outcome

The expected outcome is a student-level web prototype that demonstrates information display, form handling, database storage and administrator record management. The project supports awareness and organisation, while physical collection, transportation and recycling remain outside the software itself.

SEMINAR and PROJECT INITIATION

Assessment / Evaluation

Formative Assessment

Sr. No.	Enrollment Number	Seat Number	Name of student
1	24610960177	388497	Tejas Dinesh Barve
2	24610960179	388499	Nikhil Manohar Bhadane
3	24610960180	388500	Kunal Sanjay Bhamare
4	24610960186	388505	Keshav Vijay Chaudhari

Sr. No.	**Criteria**	**Weightage**	**Exam Seat Number**			
1	Selection Topic / Theme of seminar	5	388497	388499	388500	388505
1	Selection of Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of preparation and innovativeness	5				
4	Q-A handling	5				
5	Time management	5				
6	Seminar presentation report	10				
7	Selection of theme of problem statement and its innovativeness	5				
8	Stages of development of action plan	5				
9	Prototyping	5				
	Total	50				

Sign :

Name : Shraddha S Hajari

(Course Expert/s)